UrbanHeat

Al Khor area study, Qatar

Theme: Urban Expansion, Land Use Change & Heat Risk

Satellite and local evidence for heat-protection site assessments

Team Al Zubarah, Qatar

UAE Space Academy hackathon, October 2026

The planning problem

Proposed user: a municipal planning team choosing where to inspect first.

Shortlist cells, then inspect use, shade, access, ownership and feasibility before choosing an intervention.

Proposed value: organise evidence and inspection decisions. No adoption or measured benefit is established.

The prototype supports investigation; it does not establish heat-health risk.

Bus-stop site assessment: V04-23

Satellite: 50.4 °C mean of two cell medians; 0% detected green pixels at the current threshold.

Local evidence: roadside trees, no shelter and limited walking access; Nasser confirms site visits.

Next check: waiting-time shade, passenger use, route safety, ownership and feasibility.

300 m September values are not stop measurements. October field photos show context, not measured use or shade.

Open satellite data and mapped context

Sentinel-2: green-pixel indicators from 5, 15 and 30 September 2026.

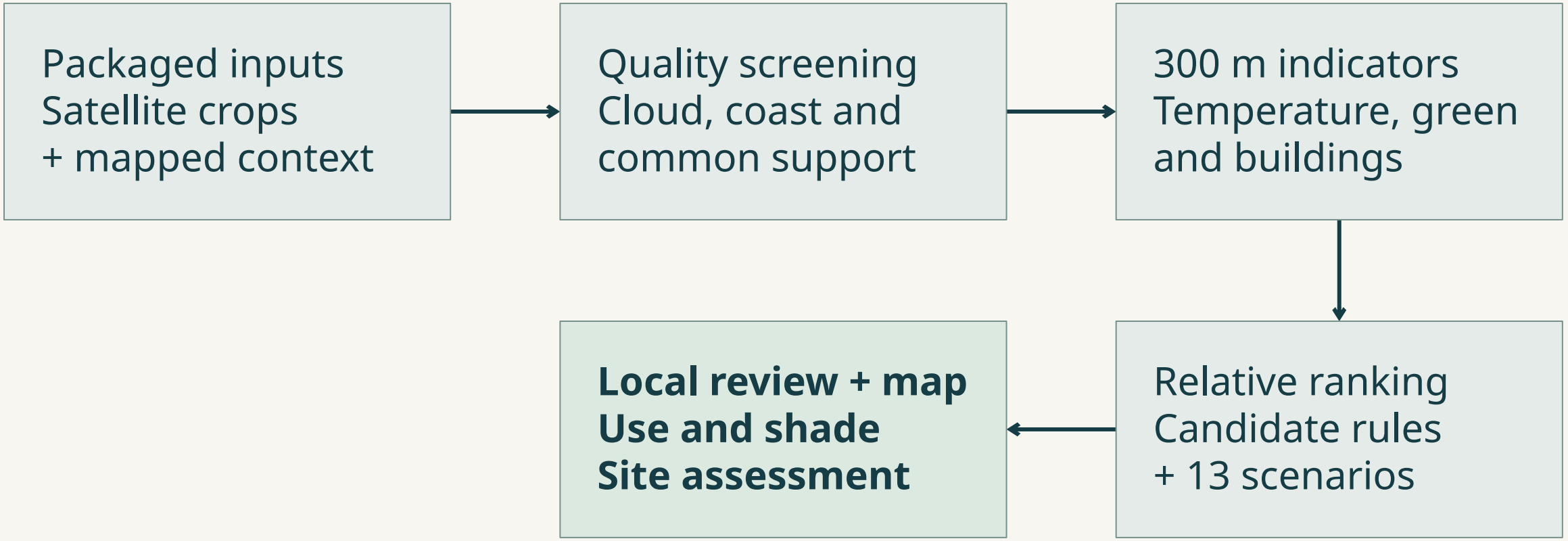
Landsat: surface temperature from 14 and 30 September. The 6 September scene fails coverage QA.

WorldCover 2021: historical land-cover context. OpenStreetMap: contributed building footprints downloaded 5 October.

Field photos: 8 October 2026; thermal data: September. No hyperspectral, 813 or proprietary VHR data.

Thermal native detail is about 100 m, delivered on a 30 m grid. We report 300 m cells.

From source crops to an inspectable shortlist



External land cover: WorldCover for heat screening; Dynamic World in a separate experiment.
QA/ranking are rule based. No new model trained.

Al Khor presentation examples

V04-23 | Bus-stop road



Trees and stop sign; no shelter visible.

V28-17 | Stadium parking



Stadium and parking; use unmeasured.

V18-26 | Jogging route



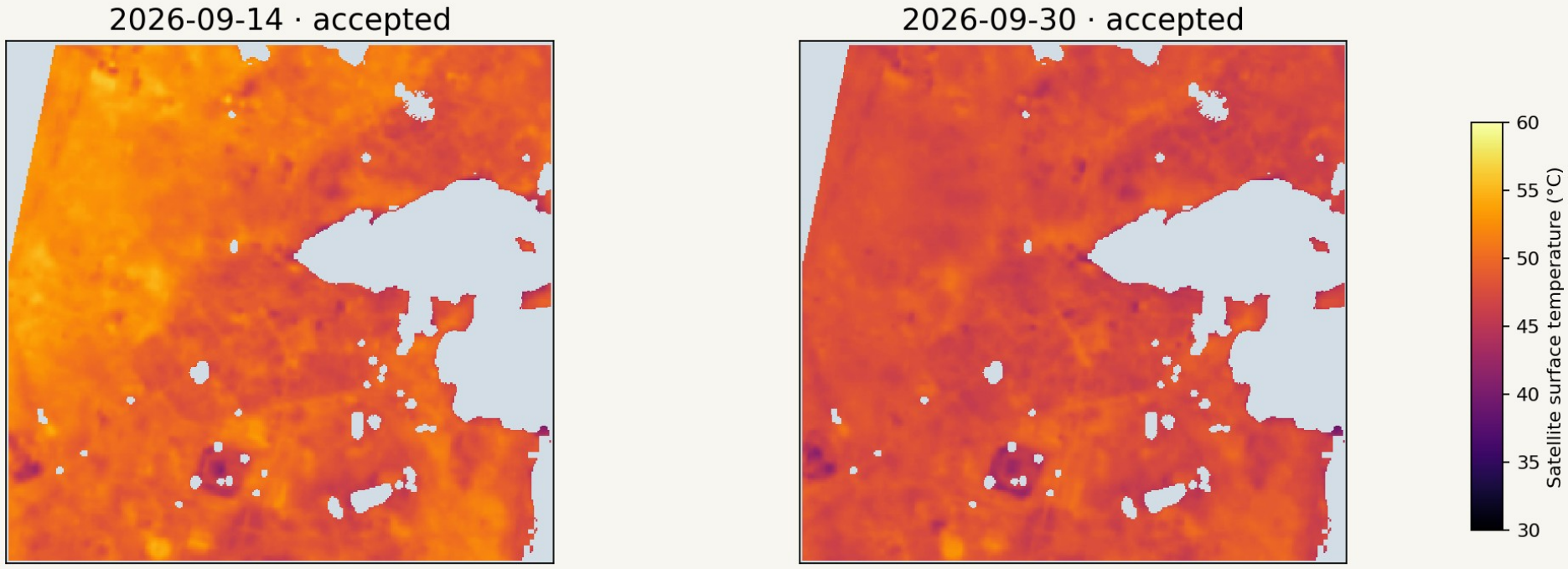
Marked path beside road and sandy ground.

8 October 2026 | 13:00–14:00 Qatar (UTC+3) | Shared capture window

Photographer: Abdulrahman Almohannadi | Confirmed by Nasser, not EXIF | Exact per-file time: Unknown

Team-selected examples; all 1,111 observed cells and 373 candidates remain. The satellite top three are unchanged.

Matched surface-temperature observations



6 September excluded: 0% of historical urban inland support passed the strict filters.
94.4379 km² of common inland sample footprints. Grey: excluded or missing observations.

Morning surface observations, not air temperature. Native thermal detail about 100 m, delivered at 30 m.
Date differences do not establish long-term cooling. Landsat imagery courtesy of USGS.

94.4379 km² of common inland sample footprints across the broad study window.

Three examples for site investigation

Case study	Surface °C*	Green %*	Buildings %	Rank range
V04-23	50.4	0.0	2.3	1-14
V28-17	50.0	0.0	0.1	1-23
V18-26	47.47	0.0386	0.0	203–336

Bus stop: waiting-area shade. Stadium: event use. Jogging route: reported lack of path shade. Photos received; use and detailed shade unmeasured.

Team-selected examples, not the top three ranks. V18-26 ranks 275/373; hospital land stays in the full results. Visits do not measure use or shade.

*Means of date-level cell summaries. V18-26: 48.02/46.92°C; 72% coverage. Scenario ranges are not confidence intervals.

Reproducibility and reference checks

The package includes 35 hashed input files, source metadata and pinned Python 3.12 dependencies.

Clean GitHub execution passed through a normal Jupyter kernel: five cells, zero errors and all 35 input hashes verified.

NDVI >=0.30 at 24/25 grass references and 0/30 building references. All 30 coastal references are excluded from retained thermal support.

Contributed reference labels have conflicts. Human confirmation and independent thermal validation remain pending.

Potential impact and incubation next steps

Completed: satellite screening, sensitivity checks, local examples and preliminary review of 18 sites across four dates.

Proposed pilot: planner feedback and field shade/use/access checks; evaluate shortlist usefulness and feasibility.

Limited Dynamic World experiment: common coverage 30.94% (50% gate). Independent scientific validation pending; 2023 deferred.

The current PoC does not measure urban expansion, health risk, causal cooling or long-term warming.

Team responsibilities

Nasser Alhuri: team coordination, site-visit confirmation and submission.

Abdulrahman Almohannadi: supplied 11 photos + one video; project review remains pending.

Mohammed Almarri: rehearse the live pitch (up to 10 min) and 5 min Q&A.

Majed Alkuwari: review slides and explain the limitations.

Ali Alkubaisi: rehearse the offline demo and prepare five-minute judge Q&A.

Photographed-site visits and supplied media collection confirmed by Nasser. Detailed assessment, member reviews and rehearsals remain pending.

Data attribution and sources

Actual sources: Copernicus Sentinel, Landsat (USGS), WorldCover 2021, OSM; separate experimental Dynamic World exports (2021, 2026).

Credits: modified Copernicus Sentinel data (2026); Landsat courtesy USGS; WorldCover CC BY 4.0; OSM ODbL.

Dynamic World, CC BY 4.0: This dataset is produced for the Dynamic World Project by Google in partnership with National Geographic Society and the World Resources Institute.

Field photos: Abdulrahman Almohannadi, 8 October 2026, shared 13:00–14:00 Qatar (UTC+3) window. Confirmed by Nasser, not EXIF; exact per-file times Unknown.

Modified Copernicus Sentinel data (2021, 2026). Full sources, licences and experimental limits accompany the package. Code: MIT.